Eligibility Traces

Reinforcement Learning

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1 Setting and Notation

We consider the *prediction* problem in model-free reinforcement learning: given a fixed policy π, estimate its state-value function from sampled trajectories generated by following π. A single trajectory has the form

$$S_1, A_1, R_2, S_2, A_2, R_3, \cdots \sim \pi.$$

The discounted return from time t is

$$G_t := \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots, \qquad 0 \le \gamma < 1.$$

Our goal is to estimate the state-value function

$$V^\pi(s) := \mathbb{E}^\pi[G_t \mid S_t = s], \qquad \forall\, s \in \mathcal{S}.$$

2 A Spectrum of Prediction Methods

2.1 Monte Carlo

For episodic tasks with terminal time T, the full return from time t is

$$G_t^{(\infty)} := R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T.$$

The Monte Carlo idea is to run the episode to completion, then use this realized return as the update target for every visited state. The estimate is unbiased but requires a complete episode before any update can be made, and has high variance because the target depends on the entire trajectory from t to T.

2.2 TD(0)

Temporal-difference learning avoids waiting until the episode ends. Instead of the full return, TD(0) uses a one-step bootstrapped target

$$G_t^{(1)} := R_{t+1} + \gamma V^\pi(S_{t+1}),$$

and defines the one-step TD error

$$\delta_t := R_{t+1} + \gamma V^\pi(S_{t+1}) - V^\pi(S_t).$$

This allows an update after every single transition, at the cost of introducing bias through the bootstrap from $V^\pi(S_{t+1})$.

2.3 n-Step TD

Monte Carlo and TD(0) are the two extremes of a natural family. The n-step return samples the first n rewards exactly and then bootstraps the remaining tail:

$$G_t^{(n)} := R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n V^\pi(S_{t+n}).$$

Setting $n = 1$ recovers TD(0). Letting $n \to \infty$ recovers Monte Carlo.

2.4 TD(λ): Combining All n-Step Returns

Rather than committing to a single n, TD(λ) forms a geometrically weighted average of all n-step returns:

$$G_t^\lambda := (1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1}G_t^{(n)}, \qquad 0 \le \lambda \le 1.$$

The weights $(1-\lambda)\lambda^{n-1}$ sum to one, so G_t^λ is a proper convex combination. The two limits are

$$\lambda = 0 \implies G_t^\lambda = G_t^{(1)} \quad \text{(TD(0))}, \qquad \lambda \to 1 \implies G_t^\lambda \to G_t^{(\infty)} \quad \text{(Monte Carlo)}.$$

As we will show in the next section, larger λ places more weight on longer-horizon returns and brings the target closer to the full Monte Carlo return.

2.5 Roadmap

The remainder of this report develops TD(λ) in detail. Section 3 proves the key identity, which re-expresses the forward-view correction $G_t^\lambda - V(S_t)$ as a geometrically weighted sum of future TD errors. Section 4 interprets this identity intuitively using the contribution table. Section 5 derives the backward view by reading the same table column-wise, and Section 6 develops eligibility traces as the efficient online mechanism for the backward view.

3 The Forward View: Key Identity

3.1 Statement

Theorem 3.1 (Forward-View Identity)**.** *Let $V : \mathcal{S} \to \mathbb{R}$ be any value estimate. Define the one-step TD error*

$$\delta_t := R_{t+1} + \gamma V(S_{t+1}) - V(S_t).$$

Then, for any $t \ge 0$,

$$G_t^\lambda - V(S_t) = \sum_{k=0}^{\infty}(\gamma\lambda)^k\delta_{t+k}.$$

3.2 Proof

Write the geometric mixture and expand each $G_t^{(n)}$ explicitly:

$$\begin{aligned} G_t^\lambda - V(S_t) = (1-\lambda)\Bigg[\quad & R_{t+1} + \gamma V(S_{t+1}) \\ & + \lambda\big(R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})\big) \\ & + \lambda^2\big(R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 V(S_{t+3})\big) \\ & + \lambda^3\big(R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \gamma^4 V(S_{t+4})\big) \\ & + \cdots \Bigg] - V(S_t). \end{aligned}$$

Collecting the coefficients of each reward term and each bootstrap value:

$$G_t^\lambda - V(S_t) = (1-\lambda)\Bigg[R_{t+1}(1+\lambda+\lambda^2+\cdots) + \gamma R_{t+2}(\lambda+\lambda^2+\lambda^3+\cdots) + \gamma^2 R_{t+3}(\lambda^2+\lambda^3+\cdots) + \cdots$$
$$+\ \gamma V(S_{t+1}) + \gamma^2\lambda V(S_{t+2}) + \gamma^3\lambda^2 V(S_{t+3}) + \cdots\Bigg] - V(S_t).$$

Evaluating each geometric series: $(1-\lambda)(1+\lambda+\lambda^2+\cdots) = 1$, $(1-\lambda)(\lambda+\lambda^2+\cdots) = \lambda$, and in general $(1-\lambda)(\lambda^j+\lambda^{j+1}+\cdots) = \lambda^j$:

$$\begin{aligned}
G_t^\lambda - V(S_t) = {} & R_{t+1} + (\gamma\lambda)R_{t+2} + (\gamma\lambda)^2 R_{t+3} + \cdots \\
& +\ \gamma V(S_{t+1}) + (\gamma\lambda)\gamma V(S_{t+2}) + (\gamma\lambda)^2\gamma V(S_{t+3}) + \cdots \\
& -\ V(S_t) - (\gamma\lambda)V(S_{t+1}) - (\gamma\lambda)^2 V(S_{t+2}) - \cdots \\
& +\ (\gamma\lambda)V(S_{t+1}) + (\gamma\lambda)^2 V(S_{t+2}) + (\gamma\lambda)^3 V(S_{t+3}) + \cdots \\
& -\ (\gamma\lambda)V(S_{t+1}) - (\gamma\lambda)^2 V(S_{t+2}) - (\gamma\lambda)^3 V(S_{t+3}) - \cdots
\end{aligned}$$

The last two lines cancel. Grouping the remaining terms into TD errors $\delta_{t+k} = R_{t+k+1} + \gamma V(S_{t+k+1}) - V(S_{t+k})$:

$$\begin{aligned}
G_t^\lambda - V(S_t) = {} & \underbrace{\big(R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\big)}_{\delta_t} \\
& +\ (\gamma\lambda)\underbrace{\big(R_{t+2} + \gamma V(S_{t+2}) - V(S_{t+1})\big)}_{\delta_{t+1}} \\
& +\ (\gamma\lambda)^2\underbrace{\big(R_{t+3} + \gamma V(S_{t+3}) - V(S_{t+2})\big)}_{\delta_{t+2}} \\
& + \cdots
\end{aligned}$$

Key Result

$$G_t^\lambda - V(S_t) = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}.$$

The forward-view correction equals a geometrically weighted sum of all future one-step TD errors, with weights $(\gamma\lambda)^k$ decaying at rate $\gamma\lambda$ per step.

This completes the proof. □

4 Intuition for the Key Identity

4.1 Why Future Errors Correct the Present Estimate

The forward-view update for $V(S_t)$ has the form

$$V(S_t) \leftarrow V(S_t) + \alpha\big(G_t^\lambda - V(S_t)\big),$$

so everything reduces to understanding the correction $G_t^\lambda - V(S_t)$. The key identity says this correction is

$$G_t^\lambda - V(S_t) = \delta_t + \gamma\lambda\,\delta_{t+1} + (\gamma\lambda)^2\,\delta_{t+2} + \cdots$$

It is not a single number available at time t. It is a sum of future TD errors. To understand why, we need to unpack the chain of bootstrap dependencies between consecutive value estimates.

Our one-step estimate for $V(S_t)$ is

$$V(S_t) \;\approx\; R_{t+1} + \gamma V(S_{t+1}).$$

The reward R_{t+1} is observed and exact. The only source of error is $V(S_{t+1})$, which is itself an estimate. The TD error δ_t measures the one-step Bellman inconsistency at S_t: by how much does $V(S_t)$ disagree with one step of lookahead?

But $V(S_{t+1})$ is not the ground truth. It is itself estimated via

$$V(S_{t+1}) \;\approx\; R_{t+2} + \gamma V(S_{t+2}),$$

and δ_{t+1} measures how wrong that estimate is. Since our estimate of $V(S_t)$ was built on $V(S_{t+1})$, any error in $V(S_{t+1})$ feeds back into our estimate of $V(S_t)$, discounted by $\gamma\lambda$. One step further: δ_{t+2} measures the error in $V(S_{t+2})$, which contaminates $V(S_{t+1})$, which in turn contaminates $V(S_t)$, arriving with weight $(\gamma\lambda)^2$. The key identity says the total correction to $V(S_t)$ is the accumulation of all these layered contributions.

> *$V(S_t)$ is estimated by bootstrapping from $V(S_{t+1})$, which is estimated by bootstrapping from $V(S_{t+2})$, and so on. Errors in any future value estimate therefore induce a correction in $V(S_t)$. The key identity makes this chain of dependencies precise: the correction at S_t is exactly the discounted accumulation of all future Bellman inconsistencies, with $(\gamma\lambda)^k$ controlling how far into the future we trust those corrections.*

4.2 The Contribution Table

It is useful to lay out all these coefficients explicitly. Each row corresponds to a state, each column to a TD error, and entry (τ, t) is the coefficient with which δ_t contributes to the update of $V(S_\tau)$, as given directly by the key identity.

	δ_1	δ_2	δ_3	δ_4	δ_5	$\delta_6 \cdots$
$V(S_1)$	1	$\gamma\lambda$	$(\gamma\lambda)^2$	$(\gamma\lambda)^3$	$(\gamma\lambda)^4$	$(\gamma\lambda)^5 \cdots$
$V(S_2)$	0	1	$\gamma\lambda$	$(\gamma\lambda)^2$	$(\gamma\lambda)^3$	$(\gamma\lambda)^4 \cdots$
$V(S_3)$	0	0	1	$\gamma\lambda$	$(\gamma\lambda)^2$	$(\gamma\lambda)^3 \cdots$
$V(S_4)$	0	0	0	1	$\gamma\lambda$	$(\gamma\lambda)^2 \cdots$
$V(S_5)$	0	0	0	0	1	$\gamma\lambda \cdots$
$V(S_6)$	0	0	0	0	0	$1 \cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 1: Coefficient with which δ_t contributes to the update of $V(S_\tau)$. Entry (τ, t) equals $(\gamma\lambda)^{t-\tau}$ when $t \geq \tau$, and 0 when $t < \tau$. The diagonal is all 1: a state always receives its own TD error in full.

Each row is the forward-view identity written out for that state. Row $V(S_3)$, for instance, reads: update $V(S_3)$ using δ_3 with weight 1, then δ_4 with weight $\gamma\lambda$, then δ_5 with weight $(\gamma\lambda)^2$, and so on. The zeros to the left of the diagonal reflect the fact that $V(S_3)$ has no dependence on δ_1 or δ_2, which were observed before S_3 was visited.

4.3 The Forward-View Algorithm

The forward view at each time t proceeds as follows:

1. Run the episode to completion and observe the full trajectory from S_t onward.
2. Compute all future TD errors $\delta_t, \delta_{t+1}, \delta_{t+2}, \ldots$
3. Form the weighted sum $\sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}$.
4. Apply a single update to $V(S_t)$.
5. Repeat for every visited state $t = 1, 2, \ldots, T$.

This is the row-wise computation in Table 1: for each row, scan all columns to the right of the diagonal and accumulate. Because every update requires future information, the forward view is inherently offline. No update to any $V(S_t)$ can be made until the episode ends.

4.4 The Role of λ

With the algorithm in hand, we can see precisely what λ controls. The weight of δ_{t+k} in the update to $V(S_t)$ is

$$(\gamma\lambda)^k = \underbrace{\gamma^k}_{\text{MDP discounting}} \cdot \underbrace{\lambda^k}_{\text{algorithmic trust}} .$$

These two factors are conceptually independent. The factor γ^k is the standard MDP time-discount: corrections further in the future are intrinsically worth less. The factor λ^k is a separate algorithmic design choice controlling how far along the chain of bootstrap dependencies we trust. It is not imposed by the MDP.

Setting $\lambda = 0$ zeroes out all weights for $k \geq 1$, leaving only δ_t. We trust nothing beyond the immediate one-step error. This recovers TD(0). Setting $\lambda \to 1$ makes $(\gamma\lambda)^k \to \gamma^k$, so all future errors contribute with the full MDP discount. This approaches Monte Carlo, which uses the complete return and does not truncate the chain of dependencies at any finite horizon.

5 From Forward View to Backward View: The Column Perspective

5.1 The Click: From Offline to Online

Return to Table 1 and focus on a single entry: the coefficient $(\gamma\lambda)^2$ sitting at row $V(S_1)$, column δ_3.

Ask yourself: at time $t = 3$, the moment δ_3 is computed, do we already know this coefficient? Yes, completely. It is $(\gamma\lambda)^{3-1} = (\gamma\lambda)^2$, determined entirely by the fact that S_1 was visited at time 1 and we are currently at time 3. It does not depend on δ_4, δ_5, or anything that happens in the future.

Now look at the entire column δ_3. Every entry in that column is known at time $t = 3$:

- $V(S_1)$ should receive δ_3 with weight $(\gamma\lambda)^2$, because S_1 was visited 2 steps ago.
- $V(S_2)$ should receive δ_3 with weight $(\gamma\lambda)^1 = \gamma\lambda$, because S_2 was visited 1 step ago.
- $V(S_3)$ should receive δ_3 with weight $(\gamma\lambda)^0 = 1$, because S_3 is the current state.
- $V(S_4), V(S_5), \ldots$ receive weight 0, because they have not been visited yet.

Every number in column δ_3 is determined at time $t = 3$.

This is the central observation. The forward view processes the table row by row: for each state S_τ, scan rightward and wait for all future TD errors. But the entries in the table are not determined by the future. The coefficient $(\gamma\lambda)^{t-\tau}$ depends only on how long ago S_τ was visited. No future information is involved at all.

The forward view is waiting for something it does not actually need to wait for.

The alternative strategy. Instead of processing the table row by row, process it column by column. At time t, the moment δ_t is observed, apply its contribution to every previously visited state immediately:

$$V(S_\tau) \leftarrow V(S_\tau) + \alpha \cdot \delta_t \cdot (\gamma\lambda)^{t-\tau}, \qquad \tau = 1, 2, \ldots, t.$$

Then at time $t + 1$, when δ_{t+1} arrives, apply column δ_{t+1} immediately. And so on, step by step throughout the episode.

By the end of the episode, what has each $V(S_\tau)$ received in total? It has received, one at a time as each TD error arrived:

$$\delta_\tau \cdot 1, \quad \delta_{\tau+1} \cdot \gamma\lambda, \quad \delta_{\tau+2} \cdot (\gamma\lambda)^2, \quad \ldots$$

The total accumulated update is

$$\sum_{k=0}^{\infty} (\gamma\lambda)^k \, \delta_{\tau+k},$$

which is exactly the forward-view correction $G_\tau^\lambda - V(S_\tau)$.

The table is the same. The numbers are the same. The total updates are identical. The only thing that changed is the order of computation: rows versus columns.

Key Result

Forward view (row-wise, offline): fix a state S_τ, wait for all future TD errors, then apply a single update:

$$V(S_\tau) \leftarrow V(S_\tau) + \alpha \sum_{k=0}^{\infty} (\gamma\lambda)^k \, \delta_{\tau+k}.$$

One state is updated. The entire future trajectory is required.

Backward view (column-wise, online): at each step t, the moment δ_t arrives, update every previously visited state immediately:

$$V(S_\tau) \leftarrow V(S_\tau) + \alpha \cdot \delta_t \cdot (\gamma\lambda)^{t-\tau}, \qquad \forall\, \tau \leq t.$$

All past states are updated. No future information is required.

The backward view is not an approximation of the forward view. It is the forward view, computed in a different order. The coefficients $(\gamma\lambda)^{t-\tau}$ are determined entirely by the past, by when each state was last visited, and are therefore available online the moment δ_t is computed. The remaining question is purely computational: how do we maintain these weights efficiently without storing the full history of the trajectory? That is what eligibility traces solve.

5.2 The Backward-View Algorithm

The backward-view algorithm at each time step t proceeds as follows:

1. Take action $A_t \sim \pi(\cdot \mid S_t)$, observe R_{t+1} and S_{t+1}.
2. Compute the current TD error:

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t).$$

3. For every state S_τ visited at time $\tau \leq t$, apply the update:

$$V(S_\tau) \leftarrow V(S_\tau) + \alpha \cdot \delta_t \cdot (\gamma\lambda)^{t-\tau}.$$

Step 3 is the column-wise operation: δ_t is distributed to every past state, weighted by recency. No future information is needed. Compare this directly with the forward view:

- Forward view: update only $V(S_t)$, using all future errors $\delta_t, \delta_{t+1}, \delta_{t+2}, \ldots$
- Backward view: update all past states $V(S_1), \ldots, V(S_t)$, using only the current error δ_t.

The roles of which state and which TD error are exactly swapped.

Step 3 as stated requires iterating over the entire visited history at every time step. If the episode has run for t steps, this costs $O(t)$ work per step, which becomes unacceptable for long episodes. We do not need to store the full history, however. For each state s, we need only maintain a single running weight: the total coefficient $(\gamma\lambda)^{t-\tau}$ accumulated from all past visits to s. This is precisely what the eligibility trace captures.

6 Eligibility Traces

6.1 Definition

At time t, the *eligibility trace* of state s is defined as

$$e_t(s) := \sum_{\tau=1}^{t} (\gamma\lambda)^{t-\tau}\, \mathbf{1}\{S_\tau = s\}, \qquad e_0(s) = 0 \;\; \forall\, s \in \mathcal{S}.$$

This is exactly the entry in column δ_t of Table 1 corresponding to state s: for every past time τ at which the trajectory visited s, we include the contribution $(\gamma\lambda)^{t-\tau}$, decaying geometrically with the number of steps elapsed since that visit.

To make this concrete, consider the trajectory at time $t = 5$. The eligibility traces are:

$$\begin{aligned}
e_5(S_1) &= (\gamma\lambda)^4, \\
e_5(S_2) &= (\gamma\lambda)^3, \\
e_5(S_3) &= (\gamma\lambda)^2, \\
e_5(S_4) &= (\gamma\lambda)^1 = \gamma\lambda, \\
e_5(S_5) &= (\gamma\lambda)^0 = 1.
\end{aligned}$$

These are exactly the entries in column δ_5 of Table 1, read top to bottom. The eligibility trace vector $\{e_t(s)\}_{s\in\mathcal{S}}$ stores the entire current column of the table in a single vector indexed by state.

6.2 The Recursive Update

Computing $e_t(s)$ from scratch at each step via the definition above requires iterating over the full history at cost $O(t)$. We now derive a one-step recursion that reduces this to $O(|\mathcal{S}|)$.

Peel off the $\tau = t$ term from the definition:

$$\begin{aligned}
e_t(s) &= \sum_{\tau=1}^{t} (\gamma\lambda)^{t-\tau}\, \mathbf{1}\{S_\tau = s\} \\
&= (\gamma\lambda)^0\, \mathbf{1}\{S_t = s\} + \sum_{\tau=1}^{t-1} (\gamma\lambda)^{t-\tau}\, \mathbf{1}\{S_\tau = s\} \\
&= \mathbf{1}\{S_t = s\} + (\gamma\lambda) \sum_{\tau=1}^{t-1} (\gamma\lambda)^{(t-1)-\tau}\, \mathbf{1}\{S_\tau = s\} \\
&= \mathbf{1}\{S_t = s\} + (\gamma\lambda)\, e_{t-1}(s).
\end{aligned}$$

Therefore the eligibility trace satisfies the recursion:

Key Result

$$e_t(s) = \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\}, \qquad e_0(s) = 0.$$

The interpretation in terms of Table 1 is immediate. Moving from column $t-1$ to column t: every existing entry is multiplied by $\gamma\lambda$, because every past visit is now one step older and therefore one factor of $\gamma\lambda$ more decayed; and a fresh 1 is added at the current state S_t, because S_t has just been visited. The recursion builds column t from column $t-1$ in a single pass over all states, at cost $O(|\mathcal{S}|)$ per step.

6.3 Intuition for the Trace Value

The trace $e_t(s)$ answers the question: how responsible is state s for what happens at time t? A state visited recently has high responsibility because it directly influenced the chain of value estimates that produced δ_t. A state visited long ago has low responsibility because that chain has been discounted by many factors of $\gamma\lambda$. A state never visited has zero responsibility.

When $\lambda = 0$, the trace collapses to $e_t(s) = \mathbf{1}\{S_t = s\}$: only the current state is eligible. The TD error δ_t updates only $V(S_t)$, recovering TD(0).

When $\lambda = 1$, the trace becomes $e_t(s) = \sum_{\tau=1}^{t} \gamma^{t-\tau}\mathbf{1}\{S_\tau = s\}$: every past visit contributes, weighted only by the MDP's own discount γ. All previously visited states remain eligible, approaching the full Monte Carlo update.

6.4 The Backward Update Rule

With eligibility traces in hand, the backward-view update at each time step t is:

1. Observe R_{t+1}, S_{t+1}, and compute $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$.
2. Update every trace: $e_t(s) \leftarrow \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\}$ for all $s \in \mathcal{S}$.
3. Update every value estimate: $V(s) \leftarrow V(s) + \alpha\, \delta_t\, e_t(s)$ for all $s \in \mathcal{S}$.

Steps 2 and 3 together cost $O(|\mathcal{S}|)$ per time step, independent of the episode length. In practice, states with $e_t(s) \approx 0$ receive negligible updates and can be skipped, making the algorithm efficient even for large state spaces.

6.5 Verification: Both Views Agree

We verify algebraically that the total update accumulated by the backward view equals the forward-view target. The backward view applies $\alpha\, \delta_t\, e_t(S_\tau)$ to $V(S_\tau)$ at each time t. The total update across the episode is proportional to

$$\sum_{t=\tau}^{\infty} \delta_t \cdot e_t(S_\tau).$$

Assuming S_τ is visited once at time τ, we have $e_t(S_\tau) = (\gamma\lambda)^{t-\tau}$ for all $t \geq \tau$. If S_τ is visited at multiple times, the trace accumulates a contribution from each visit, and by linearity of the update

the same argument applies term by term. Therefore:

$$\sum_{t=\tau}^{\infty} \delta_t \cdot e_t(S_\tau) = \sum_{t=\tau}^{\infty} (\gamma\lambda)^{t-\tau}\, \delta_t = \sum_{k=0}^{\infty} (\gamma\lambda)^k\, \delta_{\tau+k},$$

which is exactly the forward-view correction $G_\tau^\lambda - V(S_\tau)$ from Theorem 3.1. The two views are algebraically identical.

Key Result

Eligibility traces $e_t(s) = \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\}$ store the column weights of Table 1 online and recursively, at cost $O(|\mathcal{S}|)$ per step. The update $V(s) \leftarrow V(s) + \alpha\, \delta_t\, e_t(s)$ applies each column of the table immediately when δ_t is observed. Summing these online updates across the episode recovers exactly the forward-view target $G_\tau^\lambda - V(S_\tau)$ for every state. Backward TD(λ) is the online, efficient implementation of forward TD(λ).

7 Summary

The chain of ideas in this report is as follows.

The n-step returns interpolate between TD(0) and Monte Carlo. Taking their geometric mixture with weights $(1-\lambda)\lambda^{n-1}$ defines the λ-return G_t^λ, the target of the forward view.

The key identity (Theorem 3.1) re-expresses the forward-view correction $G_t^\lambda - V(S_t)$ as a geometrically weighted sum of all future TD errors with decay rate $\gamma\lambda$. The weight $(\gamma\lambda)^k$ factorizes into the MDP's own discounting γ^k and an algorithmic trust parameter λ^k. Setting $\lambda = 0$ recovers TD(0); setting $\lambda \to 1$ recovers Monte Carlo.

The contribution table (Table 1) makes the identity concrete: entry (τ, t) is the coefficient with which δ_t updates $V(S_\tau)$. Reading row τ gives the forward view: scan all future TD errors to the right of the diagonal and accumulate. Reading column t gives the backward view: when δ_t arrives, distribute it to all previously visited states above the diagonal, weighted by recency. The forward view is inherently offline because it requires future information. The backward view is online because all column weights are determined by the past alone.

Eligibility traces $e_t(s)$ store the column weights recursively via $e_t(s) = \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\}$, at cost $O(|\mathcal{S}|)$ per step. The update $V(s) \leftarrow V(s) + \alpha\, \delta_t\, e_t(s)$ applies each column of the table as soon as δ_t is observed. Summing these online updates across the episode recovers exactly the forward-view target for every state, confirming that backward TD(λ) is the online, efficient implementation of forward TD(λ).